Sitidiyo

Imiyoboro y'amazi

Ibyo umufundi w'amazi akeneye byose kuri sitidiyo imwe ya kontineri: amazi meza, amazi ashyushye, imiyoboro y'amazi yanduye, amagerageza, n'ibyo sitidiyo ikeneye ku ifamu.

> ***Ubusobanuro:*** *iyi nyandiko yahinduwe na mudasobwa (machine draft). Igomba gusuzumwa n'umuntu uvuga Ikinyarwanda kavukire mbere yo kuyiha abakozi. Iyo hari itandukaniro, kurikiza iy'Icyongereza.*

Status: verisiyo A2, 2026-10-28, PRELIMINARY. Nta mushyitsi ucumbika muri sitidiyo mbere y'uko septic system itsinda igenzura ryayo (D-016).

Amatiyo ya sitidiyo zombi ashyirwa kimwe, bityo iyi paji ikora kuri zombi.

English: read this page in English · Gucapa: iyi paji nka PDF · Amatiyo mu bishushanyo: urupapuro rwa sitidiyo · Igihe bikorwa: gahunda y'imirimo

Mu ncamake

- **End A (impera A)**: urukuta rufunze rw'impera, iruhande rw'igitanda. **End B (impera B)**: impera ifite inzugi za kontineri, iruhande rw'ubwiherero.
- **Back wall (urukuta rw'inyuma)**: urukuta rurerure rudafite aho gufungukira. **Front wall (urukuta rw'imbere)**: urukuta rurerure rufite urugi runyerera rw'ikirahure.
- Imyanya iri mu mbonerahamwe y'ibikoresho (fittings) ibarwa muri milimetero: uhereye ku rukuta rurangiye rwa end A (x), uhereye ku rukuta rw'inyuma rurangiye (y), no kuzamuka uhereye hasi harangiye (z).

Iki	Uko byateganyijwe
Itiyo y'amazi meza (supply pipe)	PPR PN20, mm 25 ku muyoboro mukuru (main), mm 20 imbere
Aho umuyoboro mukuru winjirira (main in)	munsi ya hasi kuri end B, ukazamuka muri sink base: stop valve (robine ifunga amazi), Y-filter, check meter (mubazi y'amazi), byose kuri SV

Iki	Uko byateganyijwe
Amazi ashyushye	heater ihita ishyushya amazi (instant heater), 5.5 kW, IPX4, ku rukuta rwa end B imbere muri douche, iruhande rwa douche ifatwa mu ntoki (hand shower); nta kigega ifite. Ishyushya hafi L 4 ku munota kugeza kuri 40 °C, bigenwa kuri ecran yayo, kuri douche gusa. Sink na lavabo bihabwa amazi akonje gusa (D-053)
Imiyoboro y'amazi yanduye (drains)	WC mm 110, douche mm 50, lavabo mm 40, sink y'igikoni mm 50 ku muyoboro wayo wihariye ujya kuri grease trap (ifata amavuta); yose ifite ubuhaname bwa 1:40 cyangwa burenzeho
Umuyoboro w'umwuka (vent)	open vent ya mm 75 izamuka kuri end B, ikarangirira mm 900 hejuru y'uruzitiro rwo ku gisenge (guardrail)
Tray yo kwiyuhagiriramo (shower tray)	1000 × 900
Umuvuduko w'amazi (pressure)	bar 2.5 kugeza 3, uturuka kuri pompe yongera umuvuduko (booster pump) y'ifamu

Ryari	Iki	Intambwe
Nyuma y'ibiti (battens, S03)	First fix: umuyoboro mukuru winjira, buri muyoboro w'amazi meza n'uw'amazi yanduye, igerageza rya bar 10	S04
Hamwe n'ubwiherero	Tray (S09), WC, heater na lavabo (S10)	S09, S10
Hamwe n'igikoni	Sink, robine n'akayunguruzo (filter)	S13

Ibiva ku ifamu

- Umuyoboro umwe w'amazi akonje ufite bar 2.5 kugeza 3, uvuye ku kigega cy'amazi cy'ifamu (storage tank) no kuri pompe yongera umuvuduko (booster pump). Amazi y'umuyoboro rusange (mains) yinjira muri icyo kigega anyuze ku mwanya w'umwuka (air gap), kugira ngo amazi y'ifamu atazigera asubira mu muyoboro rusange (D-016).
- Umuyoboro w'imyanda (soil connection) uva kuri inspection chamber iri hanze ya end B ujya kuri septic tank (fosse septique) y'ifamu, n'umuyoboro w'igikoni ujya kuri grease trap ifite soak pit (puits perdu) yayo bwite.
- Septic tank ishyirwa nibura m 50 uvuye kuri borehole (iriba ricukuwe), m 3 kugeza 7 uvuye ku nyubako no ku mbibi z'ubutaka (boundary), hepfo ya borehole (downhill), itarenze m 30 uvuye aho imodoka ivoma imyanda (tanker) igera, kandi ishyirwaho nyuma ya percolation test gusa (igerageza ry'uko ubutaka bunywa amazi).

Amazi y'ikibanza n'ingano ya septic tank kuri sitidiyo zombi · download CSV

item	value
persons	4
daily_L	600
tank_m3	2.5
septic_rs143_m3	3
septic_2012_m3	2.72
septic_uk_m3	3.08
septic_m3	3.5
chamber1_m3	2.33
chamber2_m3	1.17
pump_lpm	26

Ingingo zo guhagarara ukagenzura (hold points)

1. **Ibiti mbere.** Tangira first fix gusa iyo ibiti (S03) byamaze gufatwa kandi impapuro za hasi zamaze gushyirwaho.
2. **Igerageza rya bar 10.** Uzuza amazi mu matiyo ya PPR, ugumane bar 10 mu gihe cy'isaha imwe nta kugabanuka, ubyandike kandi ubisinyire mbere ya spray foam (S05). Foam ihisha amatiyo burundu.
3. **Mbere y'abashyitsi.** Imiyoboro y'amazi yanduye yageragejwe yuzuzwa amazi (fill-test), vent yageragejwe n'umwotsi (smoke-test), amazi ya douche yapimwe n'igipimabushyuhe, kandi septic system y'ifamu yagenzuwe.

Intambwe

S04: first fix

1. Zana umuyoboro mukuru w'amazi akonje (W1, mm 25) unyuze munsi ya hasi kuri end B, uwuzamure muri sink base unyuze kuri stop valve, Y-filter na check meter, aho amazi ashobora gufungwa nta gikoresho.
2. Nyuza amazi akonje imbere mu tubati no mu nkuta: unyuze muri partition, muri lining y'urukuta rw'inyuma no mu rukuta rwa end B, ugere kuri cistern ya WC (W4), kuri lavabo (W5) no hejuru

ku kirenge cya heater muri douche (W2). Amatiyo y'amazi akonje agume mm 300 hejuru ya hasi.

3. Nta matiyo y'amazi ashyushye: amazi ashyushye yonyine ni ayo mu itiyo yoroshye (hose) ya douche ifatwa mu ntoki, ivuye hejuru kuri heater (H1).
4. Buri muyoboro ugenda ugororotse: ukurikiye urukuta, hanyuma ukazamuka ugororotse ugera aho usohokera (outlet). Nta gice cy'itiyo kigenda kiberamye (diagonal).
5. Manika imiyoboro y'amazi yanduye mu mwanya uri munsi ya hasi (kontineri ihagaze hafi mm 700 hejuru y'ubutaka): WC mm 110, douche mm 50, lavabo mm 40, ijya kuri inspection chamber (regard); sink y'igikoni mm 50 ku muyoboro wayo wihariye ujya kuri grease trap. Nyura hagati y'ibyuma byambukiranya (cross-members) bya mm 122, ntuzigere ubinyuramo.
6. Ubuhaname (fall) bwa 1:40 cyangwa burenzeho; rodding eye (aho basukurira umuyoboro) kuri buri coude (bend); vent ya mm 75 izamuka kuri end B.
7. Kora igerageza ry'umuvuduko kuri bar 10 mu gihe cy'isaha imwe, ufotore buri rukuta, usinye uruhushya rwo gushyiraho foam (release).

S09, S10 na S13: second fix

1. S09: shyiraho tray yo kwiyuhagiriramo (1000 × 900) ku mwobo wayo usohora amazi (waste) wa mm 90, hamwe na ledge y'umubaji n'ikirahure gihamye (screen).
2. S10: manika WC ifatwa ku rukuta (wall-hung) kuri cistern yayo yoroheje ihishe mu rukuta (concealed cistern): ikadiri ya mm 80 mu rukuta rwa end B, urukuta rwongerewe mm 30 imbere (build-out), na dual-flush plate.
3. S10: manika heater ihita ishyushya amazi ya 5.5 kW (200 × 330 × 90) ku rukuta rwa end B imbere muri douche, hagati ya rail n'ikirahure, ecran yayo iri hafi m 1.3 hejuru ya hasi; uhuze amazi akonje ku kirenge cyayo na hose ya douche hejuru yayo. Nta kigega ifite, bityo ntikeneye relief valve.
4. S10: shyiraho lavabo (400 × 300) na robine yayo y'amazi akonje hamwe na bottle trap (siphon) mu gice cyo gukarabiramo (wash lobby); shyiraho na rail na douche ifatwa mu ntoki ku rukuta rwa end B.
5. S13: robine y'igikoni (amazi akonje) kuri angle valve yayo, n'akayunguruzo k'amazi yo kunywa (drinking-water filter) muri sink base, gafite robine yako bwite iruhande rwa robine y'igikoni.

Amagerageza (tests)

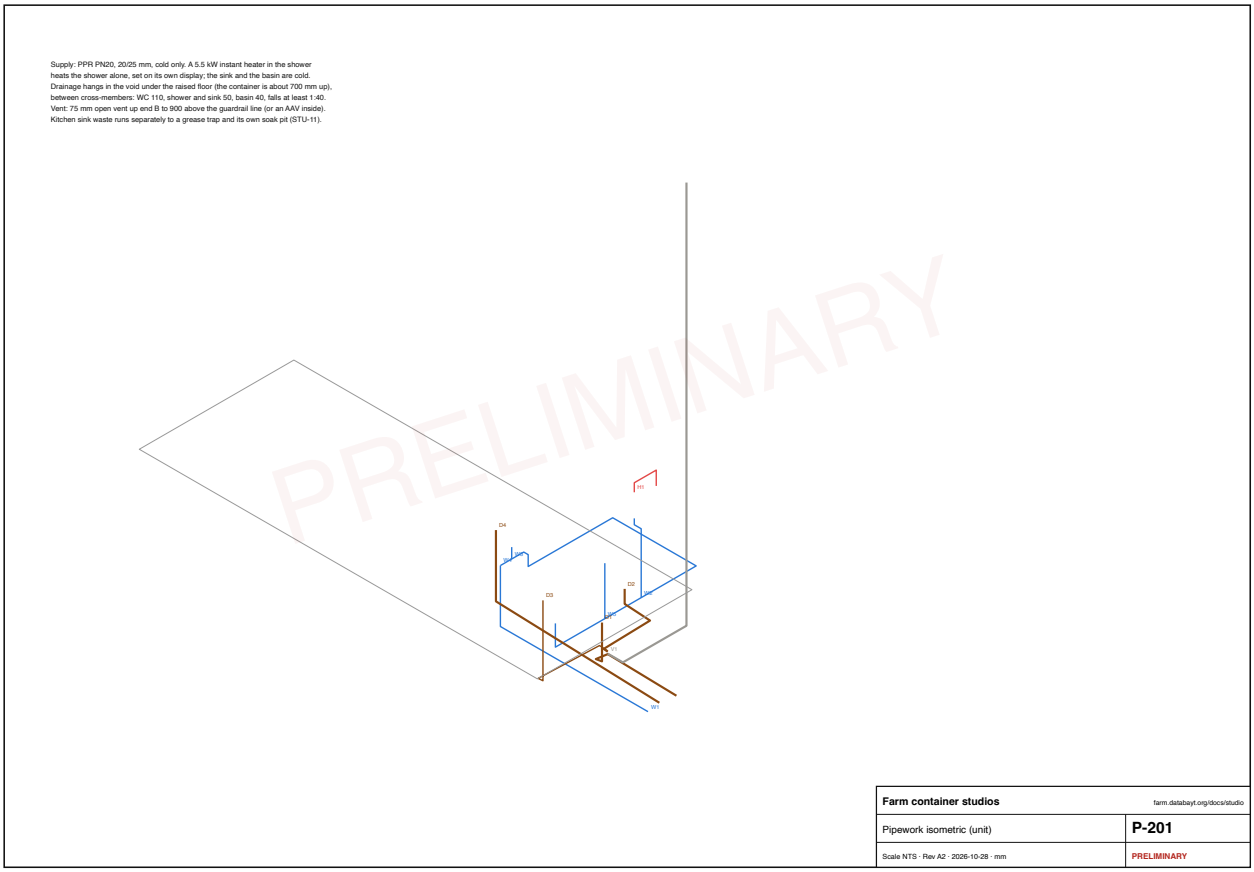
1. PPR kuri bar 10 mu gihe cy'isaha imwe, mbere ya foam (S04).
2. Gerageza buri muyoboro w'amazi yanduye uwuzuza amazi (fill-test), kandi ugenzure ko buri siphon (trap) igumana amazi ayifunga (seal).
3. Gerageza vent ukoresheje umwotsi (smoke-test).

4. Fungura douche yose, upime amazi ukoresheje igipimabushyuhe (thermometer): hafi 40 °C. Niba ashyushye cyane, gabanya ubushyuhe kuri ecran ya heater.
5. Fungura douche igihe kirekire: amazi agumana ubushyuhe, kuko heater nta kigega ifite cyashira.

Imiyoboro (runs)

Each run is checked by the model: every leg runs along a wall or straight up or down, each drain falls at least 1:40.

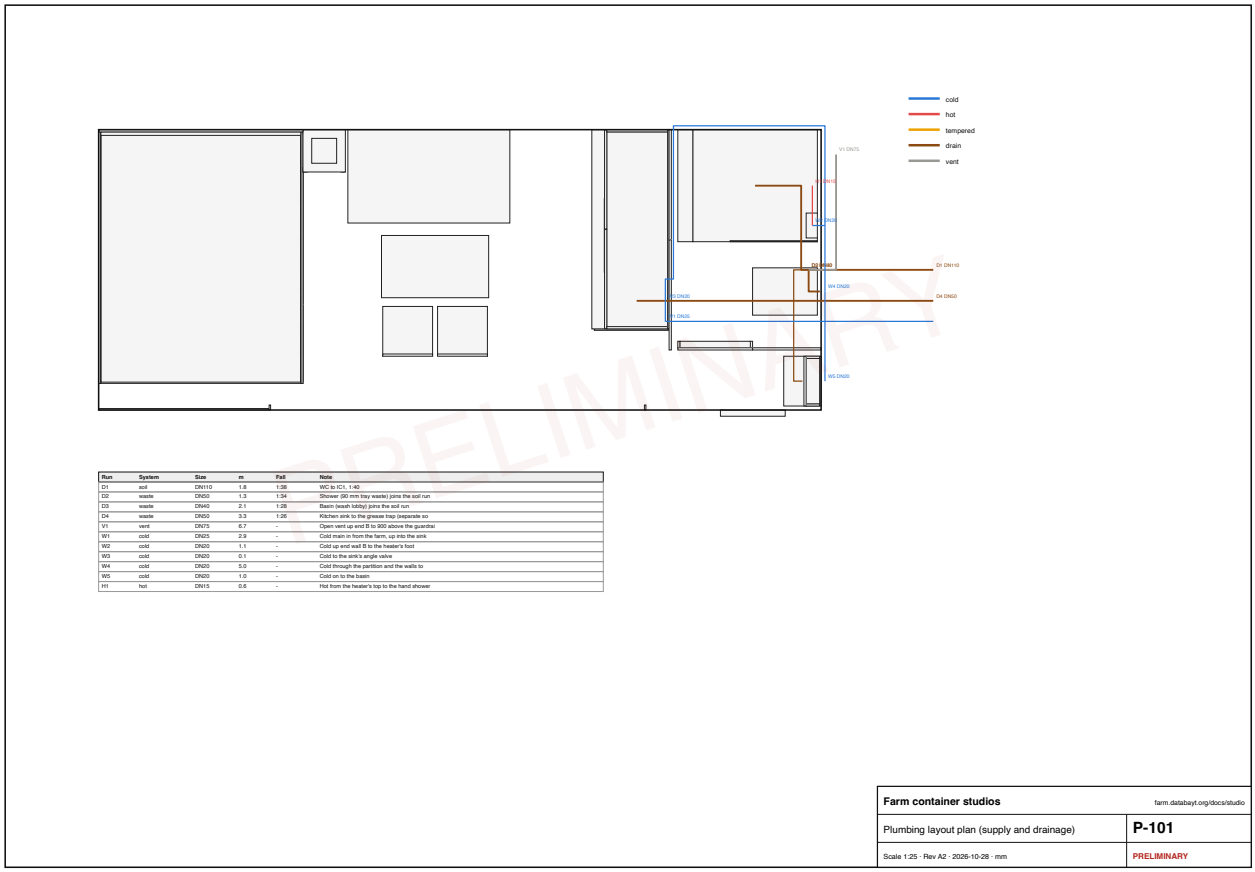
Run	System	Size	Length	Fall	What it does
D1	soil	DN110	1.8 m	1:38	WC to IC1, 1:40
D2	waste	DN50	1.3 m	1:34	Shower (90 mm tray waste) joins the soil run
D3	waste	DN40	2.1 m	1:28	Basin (wash lobby) joins the soil run
D4	waste	DN50	3.3 m	1:26	Kitchen sink to the grease trap (separate soak pit)
V1	vent	DN75	6.7 m	–	Open vent up end B to 900 above the guardrail
W1	cold	DN25	2.9 m	–	Cold main in from the farm, up into the sink base: stop valve, filter, check meter
W2	cold	DN20	1.1 m	–	Cold up end wall B to the heater's foot
W3	cold	DN20	0.1 m	–	Cold to the sink's angle valve
W4	cold	DN20	5 m	–	Cold through the partition and the walls to the WC cistern
W5	cold	DN20	1 m	–	Cold on to the basin
H1	hot	DN15	0.6 m	–	Hot from the heater's top to the hand shower: its own hose



P-201 · Pipework isometric (unit) · open full size

Points

10 points, from the stop valve on the main to the vent above the roof. The sink, the heater, the shower, the WC and the basin sit within about 1.5 m of each other, so the pipes stay short.



P-101 · Plumbing layout plan (supply and drainage) · open full size

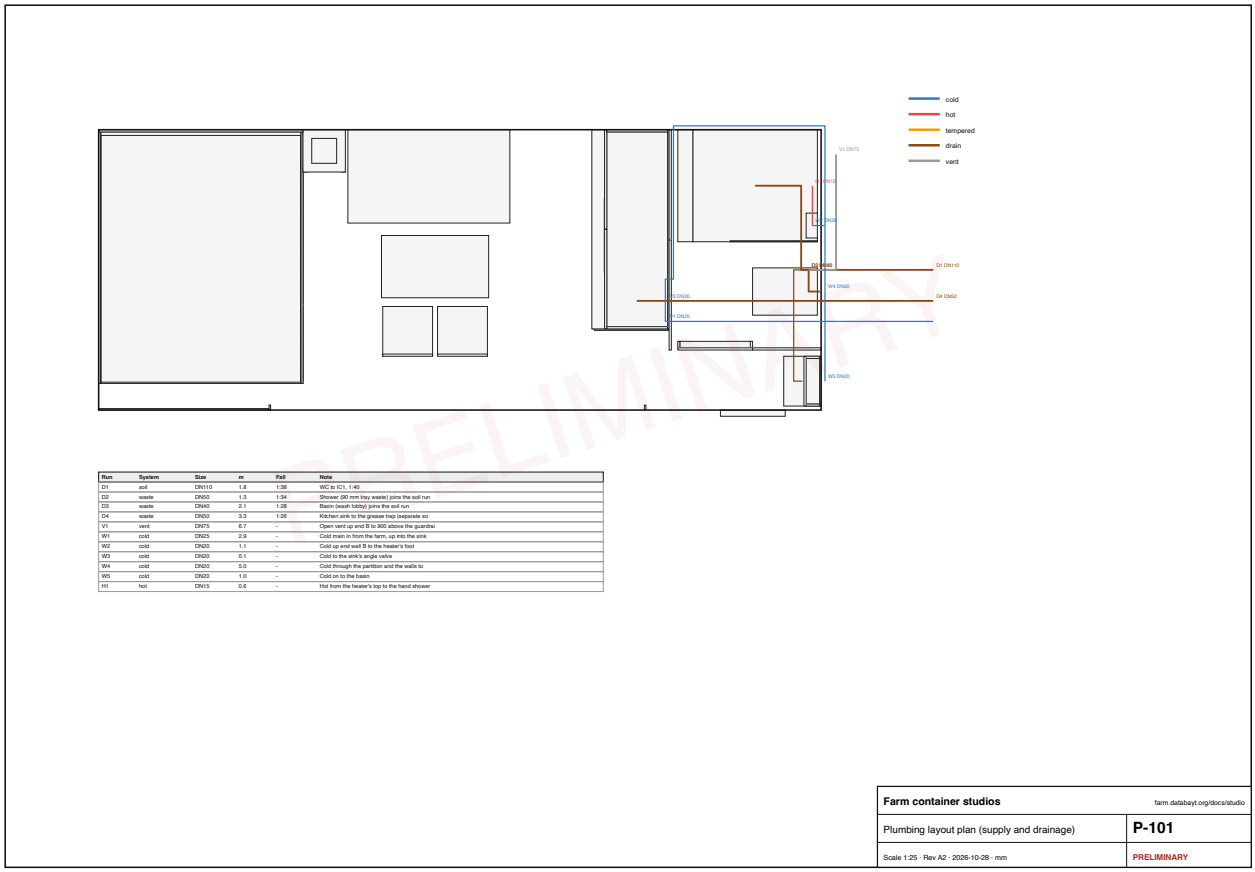
Point	What	Type	Pipes	Height	Rating
SV	Stop valve, Y-filter and check meter on the main, in the sink base	valve	W1	320 mm	DN25
WH	Instant heater on end wall B inside the shower, its display at eye height; hot only to the shower	heater	W2 H1	1,415 mm	5.5 kW, about 4 L/min, 40 °C
KT	Kitchen tap (cold), its angle valve under the sink	tap	W3 D4	900 mm	6 L/min
DF	Drinking-water filter in the sink base, its own faucet by the tap	filter	W3	350 mm	drinking water

Point	What	Type	Pipes	Height	Rating
SH	Hand shower on a slide rail on end wall B, fed by the heater's hose	shower	H1 D2	1,110 mm	about 4 L/min
WC	Concealed cistern in end wall B, dual-flush plate	WC	W4 D1	1,010 mm	dual flush
BT	Basin tap (cold), its angle valve under the console	tap	W5 D3	820 mm	6 L/min
IC1	Inspection chamber outside end B, on to the farm's septic tank	on the site	D1 D2 D3	−326 mm	DN110 in
GT	To the farm's grease trap and its own soak pit	on the site	D4	−290 mm	DN50 in
VT	Vent terminal, 900 mm above the roof guardrail	vent	V1	5,300 mm	DN75

Ibibazo bitarasubizwa

1. Septic tank y'ifamu izashyirwa he, kandi percolation test yayo yaratsinze? Imiyoboro y'amazi yanduye ya sitidiyo n'abashyitsi bayo ba mbere birategereje byombi (amazi y'ifamu).

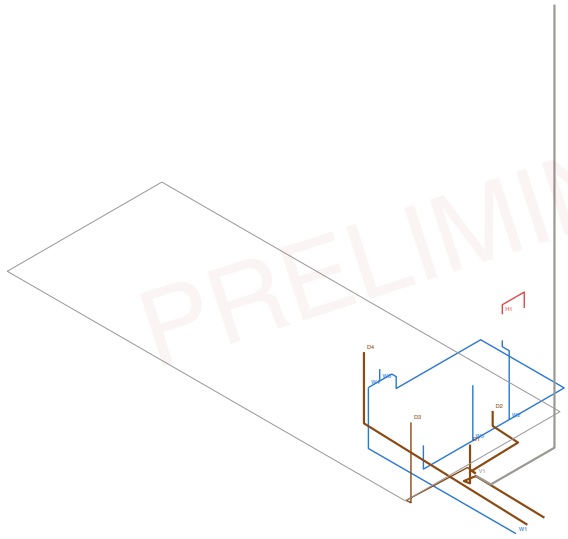
Ibishushanyo



Run	System	Size	m	Fall	Note
D1	soil	DN110	1.8	1:38	WC to IC1, 1:40
D2	waste	DN50	1.3	1:34	Shower (90 mm tray waste) joins the soil run
D3	waste	DN40	2.1	1:28	Basin (wash lobby) joins the soil run
D4	waste	DN50	3.3	1:20	Kitchen sink to the grease trap (separate so
V1	vent	DN75	6.7	-	Open vent up end B to 900 above the guardrail
W1	cold	DN25	2.9	-	Cold main in from the farm, up into the sink
W2	cold	DN20	1.1	-	Cold up end wall B to the heater's foot
W3	cold	DN20	0.1	-	Cold to the sink's angle valve
W4	cold	DN20	5.0	-	Cold through the partition and the walls to
W5	cold	DN20	1.0	-	Cold on to the basin
H1	hot	DN15	0.6	-	Hot from the heater's top to the hand shower

P-101 Plumbing layout plan · open full size

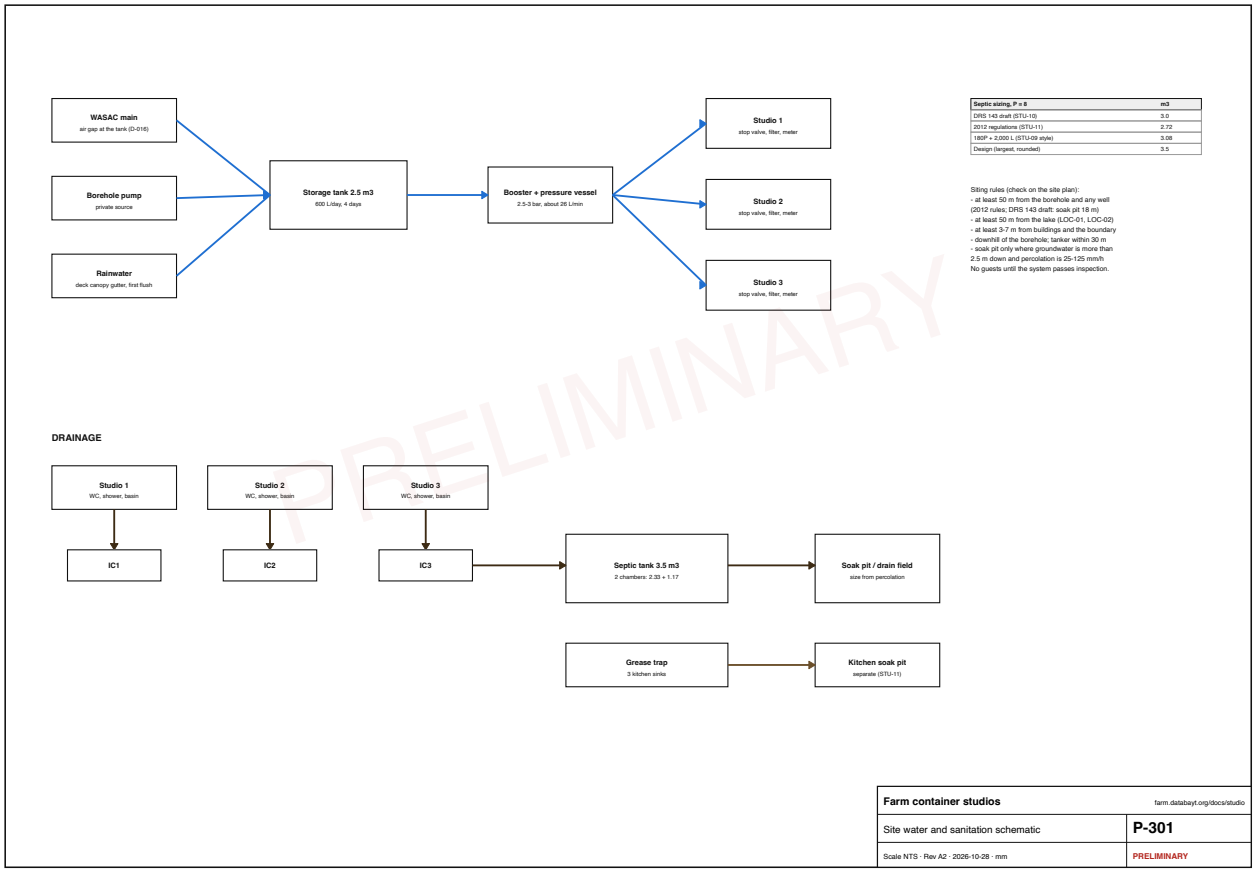
Supply: PPR PN20, 20/25 mm, cold only. A 5.5 kW instant heater in the shower heats the shower alone, set on its own display; the sink and the basin are cold.
Drainage hangs in the void under the raised floor (the container is about 700 mm up), between cross-members: WC 110, shower and sink 50, basin 40, falls at least 1:40.
Vent: 75 mm open vent up end B to 900 above the guardrail line (or an AAV inside).
Kitchen sink waste runs separately to a grease trap and its own soak pit (STU-11).



PRELIMINARY

Farm container studios	farm.databayt.org/docs/studio
Pipework isometric (unit)	**P-201**
Scale NTS · Rev A2 · 2026-10-28 · mm	PRELIMINARY

P-201 Pipework isometric · open full size



P-301 Site water and sanitation schematic · open full size